

Spatially-resolved transcriptomic data integration

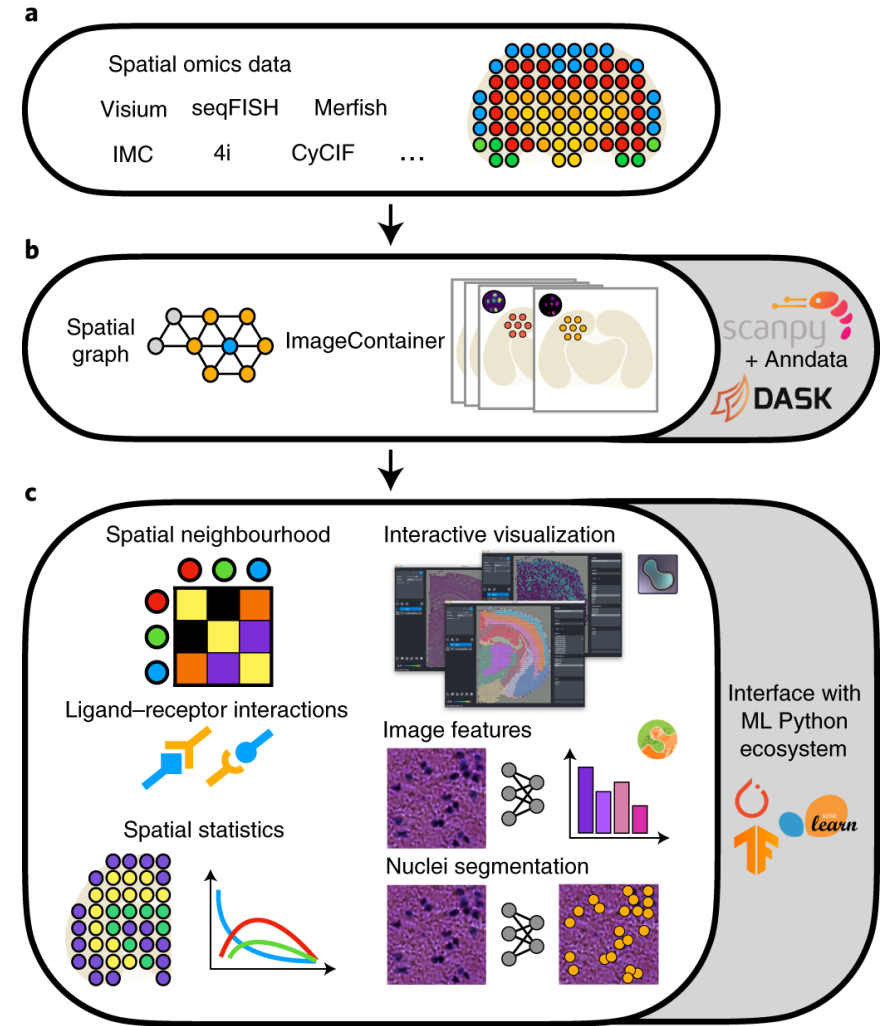
Tamim Abdelaal, PhD
Data Science, Genmab



Spatial data analysis pipelines

- Image preprocessing
- Cell segmentation / segmentation-free
- Quality control / filtering / normalization
- Downstream analysis:
 - Cell annotation
 - Spatially-aware clustering (aka. Domain identification)
 - Cell-cell communication
 - Deconvolution
 - Predicting unmeasured genes
 - Consecutive sections alignment
 - ...

Pipelines: Squidpy [Palla,...Theis, 2022], SPARROW [Pollaris*, Vanneste*,...,Saeys 2024], MCMICRO [Shapiro,..., Sorger 2021], SOPA [Balmepey,..., Cournède 2024],...



Squidpy: Palla,...Theis (2022)

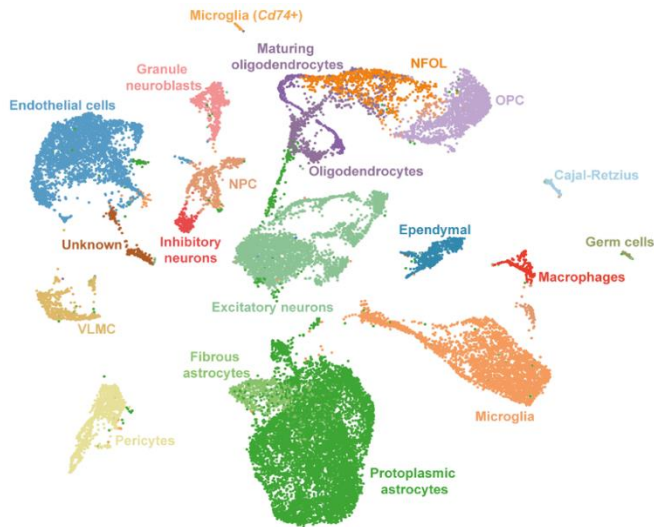
Integration challenges in SRT

- Integrating SRT and sc/snRNA-seq data
- SRT data alignment (pseudo 3D, virtual block,...)
- Spatial multiomics

Different technologies -> different questions

Single-cell/nuclei RNA-seq

(droplet, combinatorial indexing, plate-based)

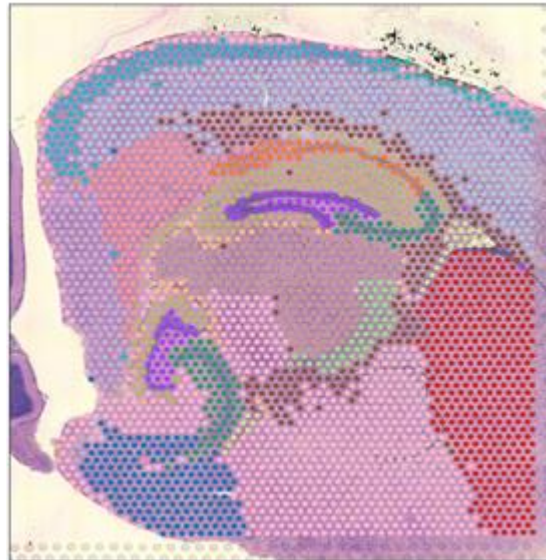


+ “Untargeted”

- No spatial information

NGS-based

(Visium, Slide-seq, Stereo-seq, Open-ST,...)

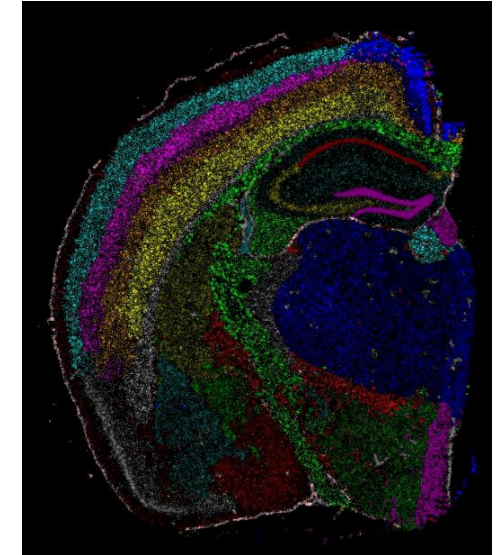


+ “Untargeted”

- Low resolution (spots)

In-situ sequencing / x-FISH

(Xenium, EEL-FISH, MERFISH, seqFISH)

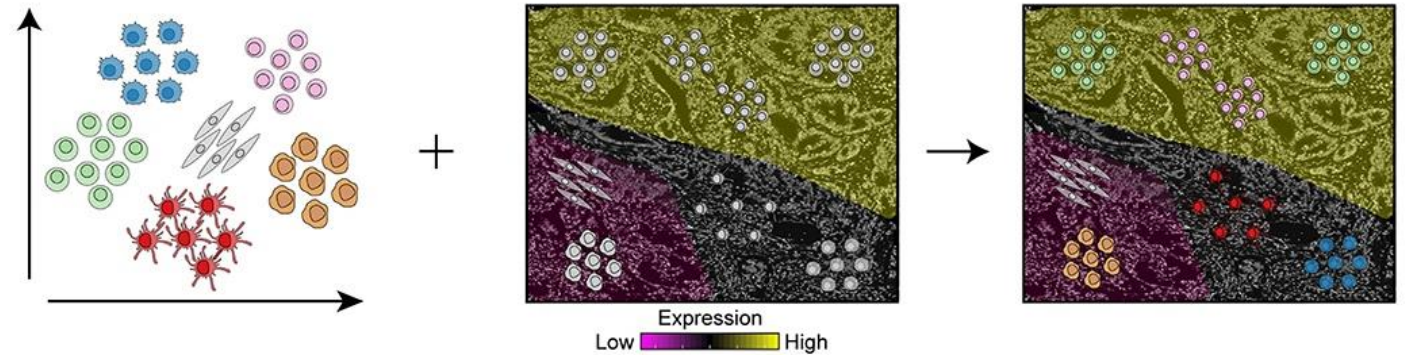


+ High resolution

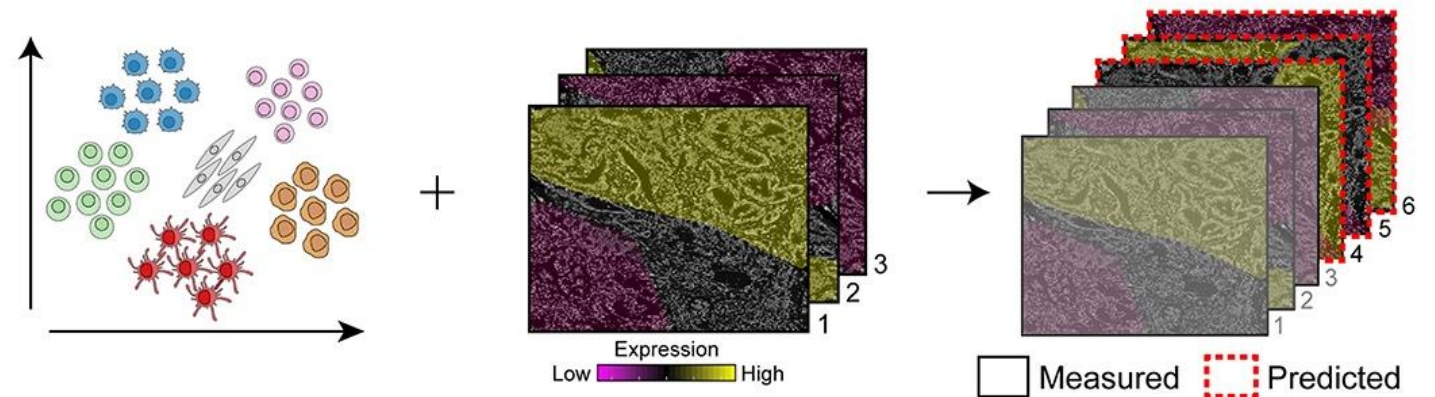
- Limited genomic features

Single-cell + spatial

- Predict the location of dissociated cells
- Deconvolution



- Predicting spatial expression of unmeasured genes



Approaches for scRNA-seq & spatial data integration

Deconvolution

Imputation

Joint embedding

e.g. SpaGE, Seurat, LIGER, gimVI, ENVI, stPlus DSTG, SD



Probabilistic modelling

e.g. CARD, cell2location, RCTD, stereoscope, SpatialDecon, STRIDE, NMFreg, SpatialDWLS, SPOTlight, Stdeconvolve, SpiceMix, Berglund



Probabilistic mapping

e.g. Tangram, novoSpaRc, SpaOTsc



Approaches for scRNA-seq & spatial data integration

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Probabilistic mapping

e.g. **Tangram**, novoSpaRc, SpaOTsc



Tangram

- Given scRNAseq ($S_{n_{cells} \times n_{genes}}$), and spatial data ($G_{n_{voxels} \times n_{genes}}$)
- Learn a mapping matrix $M_{n_{cells} \times n_{voxels}}$, $M_{ij} \geq 0$ is the probability of cell i of being in voxel j , $\sum_j^{n_{voxels}} M_{ij} = 1$.
- $M^T S$: predicted spatial gene expression, $m_j = \sum_i^{n_{cells}} \frac{M_{ij}}{n_{cells}}$: predicted cell density in voxel j

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- Objective function:

$$\Phi(\tilde{M}) = \underbrace{KL(\vec{m}, \vec{d})}_{\text{Density term}} - \underbrace{\sum_k^{n_{genes}} \cos_{sim}((M^T S)_{*,k}, G_{*,k})}_{\text{gene/voxel expression term}} - \underbrace{\sum_k^{n_{voxels}} \cos_{sim}((M^T S)_{j,*}, G_{j,*})}_{\text{voxel/gene expression term}}$$

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Tangram

- Actual objective function:

$$\begin{aligned}\Phi \left(\tilde{M}, \vec{\tilde{f}} \right) = & KL \left(\vec{\mathbf{m}}^{\mathbf{f}}, \vec{\mathbf{d}} \right) - \sum_k^{n_{genes}} \cos_{sim} \left((M^T S^f)_{*,k}, G_{*,k} \right) \\ & - \sum_j^{n_{voxels}} \cos_{sim} \left((M^T S^f)_{j,*}, G_{j,*} \right) - \lambda_{r_1} \sum_{i,j}^{n_{cells}, n_{voxels}} M_{ij} \log (M_{ij}) \\ & + abs \left(\sum_i^{n_{cells}} f_i - n_{\text{target_cells}} \right) + \sum_i^{n_{cells}} (f_i - \hat{f}_i^2).\end{aligned}$$

Tangram

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 & + abs \left(\sum_i^{n_{cells}} f_i - n_{\text{target_cells}} \right) + \sum_i^{n_{cells}} (f_i - \hat{f}_i^2).
 \end{aligned}$$

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$\vec{f_{n_{cells}}}$: filter vector

$$- \sum_j^{n_{voxels}} \cos_{sim} \left((M^T S^f)_{j,*}, G_{j,*} \right) - \lambda_{r_1} \sum_{i,j}^{n_{cells}, n_{voxels}} M_{ij} \log(M_{ij})$$

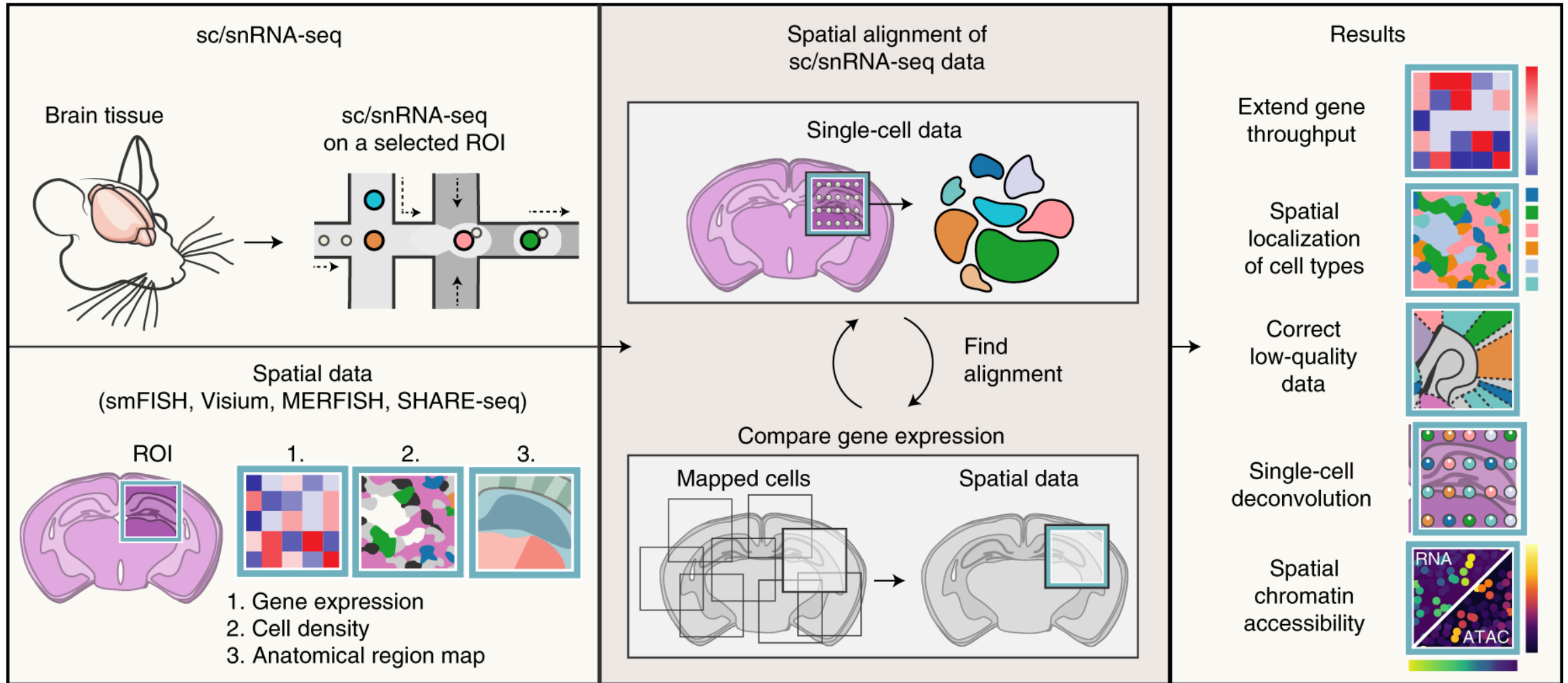
$$+ abs \left(\sum_i^{n_{cells}} f_i - n_{target_cells} \right) + \sum_i^{n_{cells}} (f_i - f_i^2).$$

Count term

Filter regularizer
(promote Boolean values)

Tangram

a



cell2location

Spatial counts $\sim NB(\mu_{s,g}, \alpha_{e,g})$

s: spatial location

g: gene

e: batch (section, slide, ...)

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$$\mu_{s,g} = \left(\underbrace{m_g}_{\text{technology sensitivity}} \cdot \underbrace{\sum_f w_{s,f} g_{f,g}}_{\text{cell type contributions}} + \underbrace{s_{e,g}}_{\text{additive shift}} \right) \cdot \underbrace{y_s}_{\text{per-location sensitivity}}$$

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Cell type abundance.
Prior: similarity in cell
type between
locations

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To account for differences between scRNA-seq and spatial

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To account for free floating RNA (background noise)

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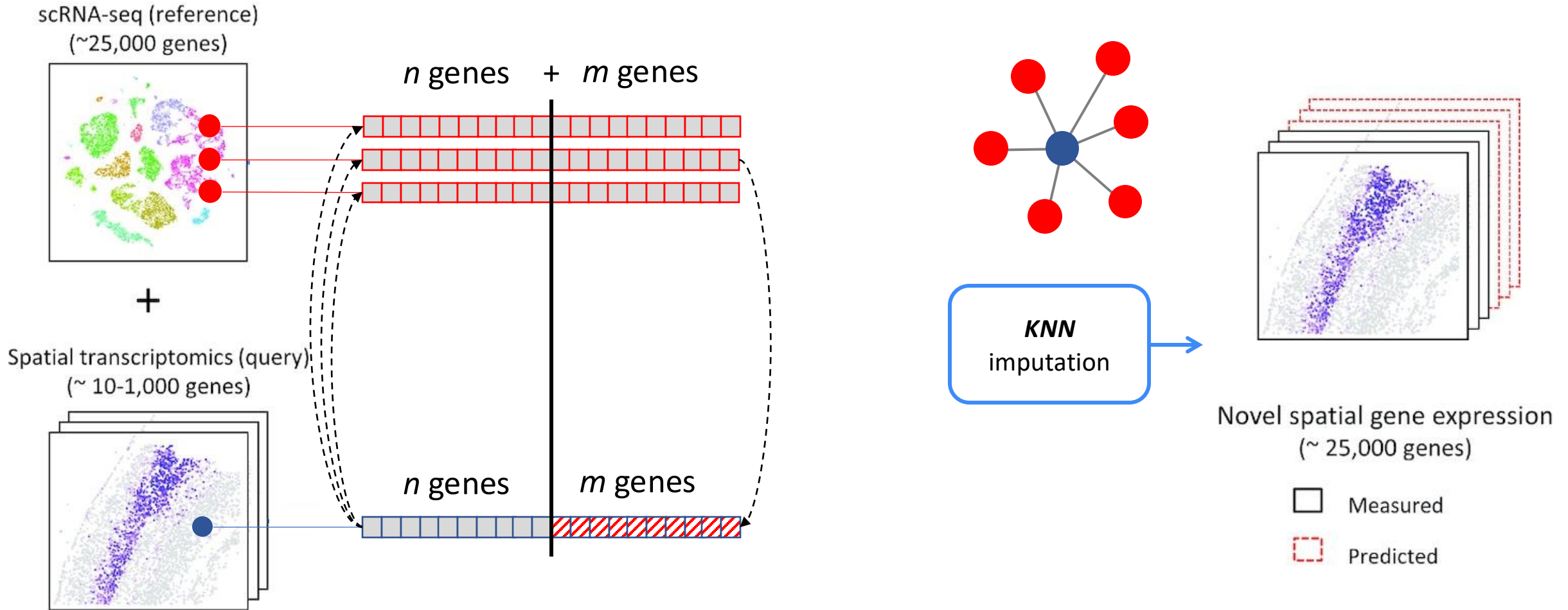
To account for differences between scRNA-seq and spatial

Cell type abundance.
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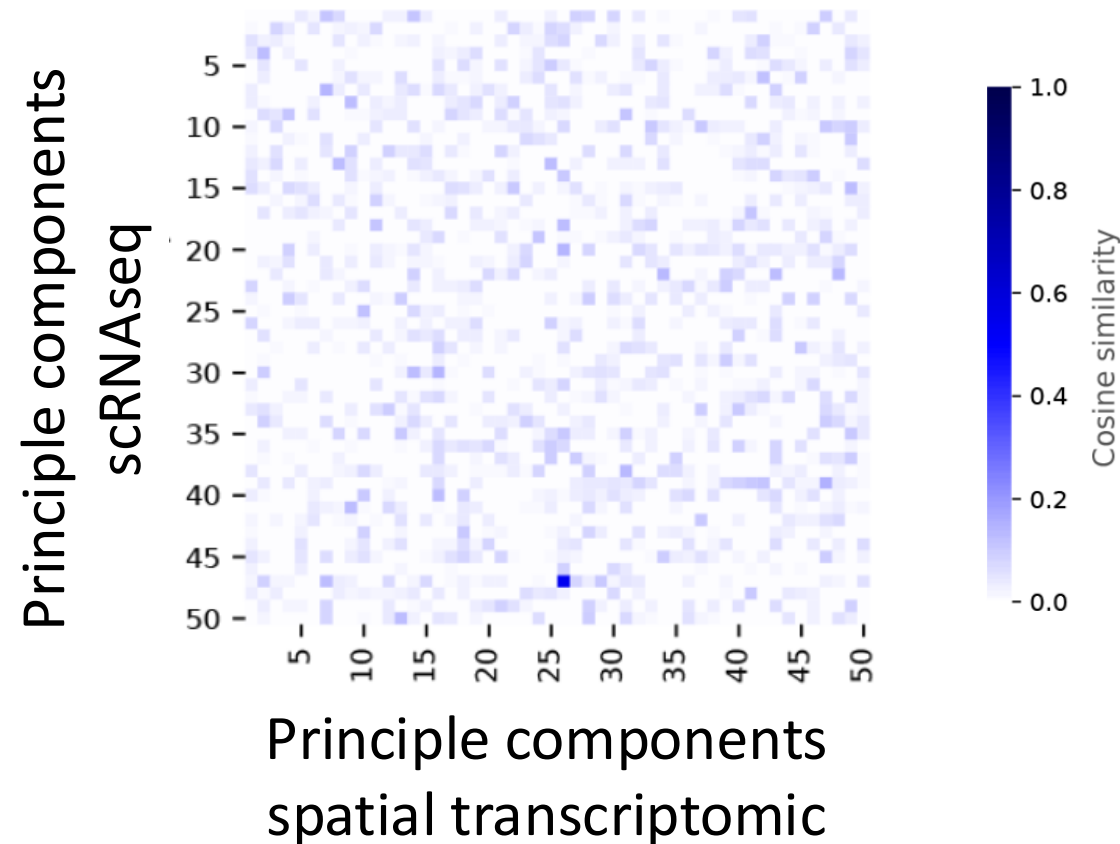
To account for differences in sensitivity within a section

SpaGE: Spatial Gene Expression Enhancement



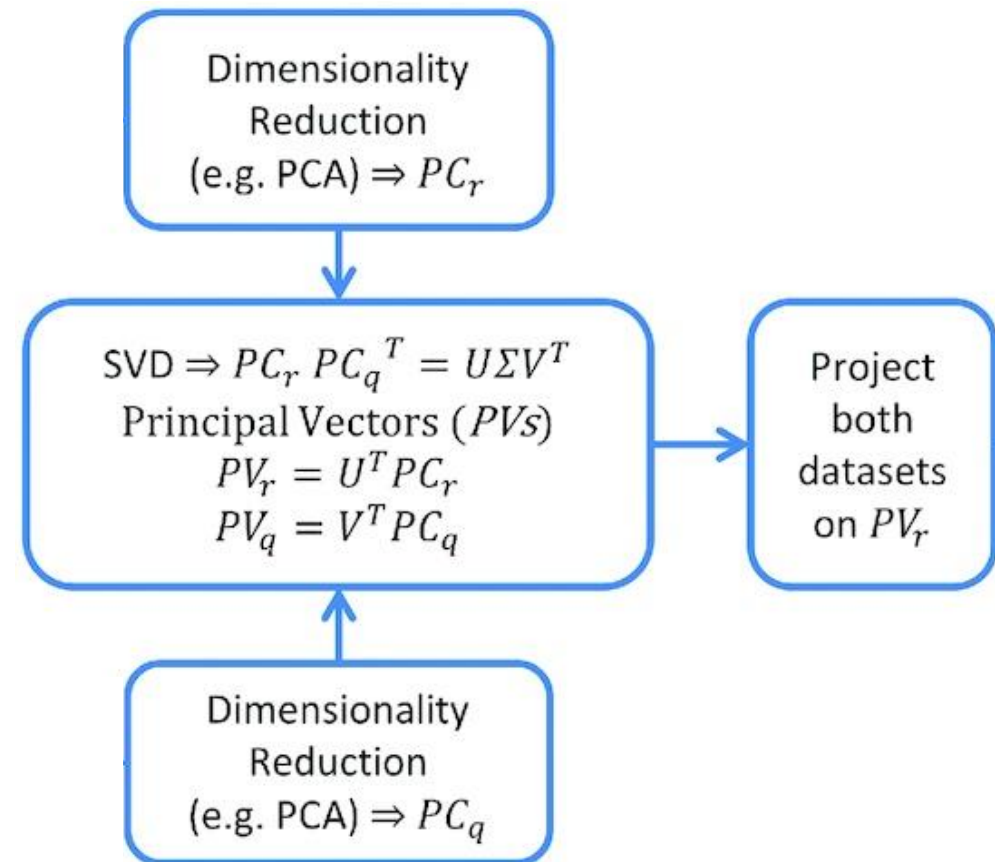
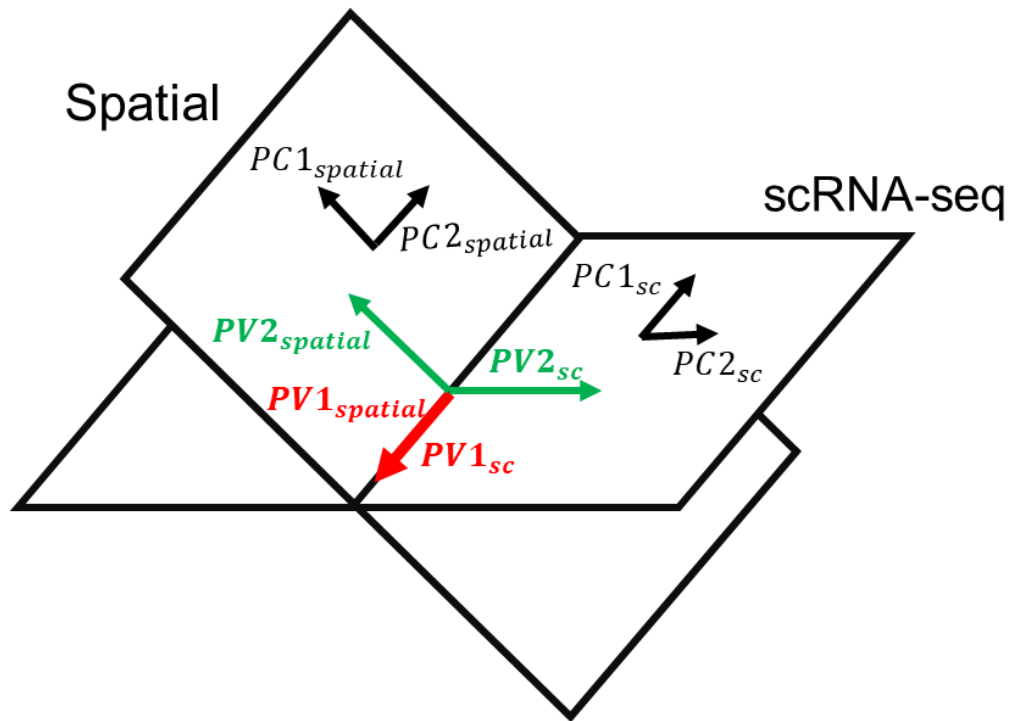
Problem: single-cell and spatial data don't align

Similarity between principal components

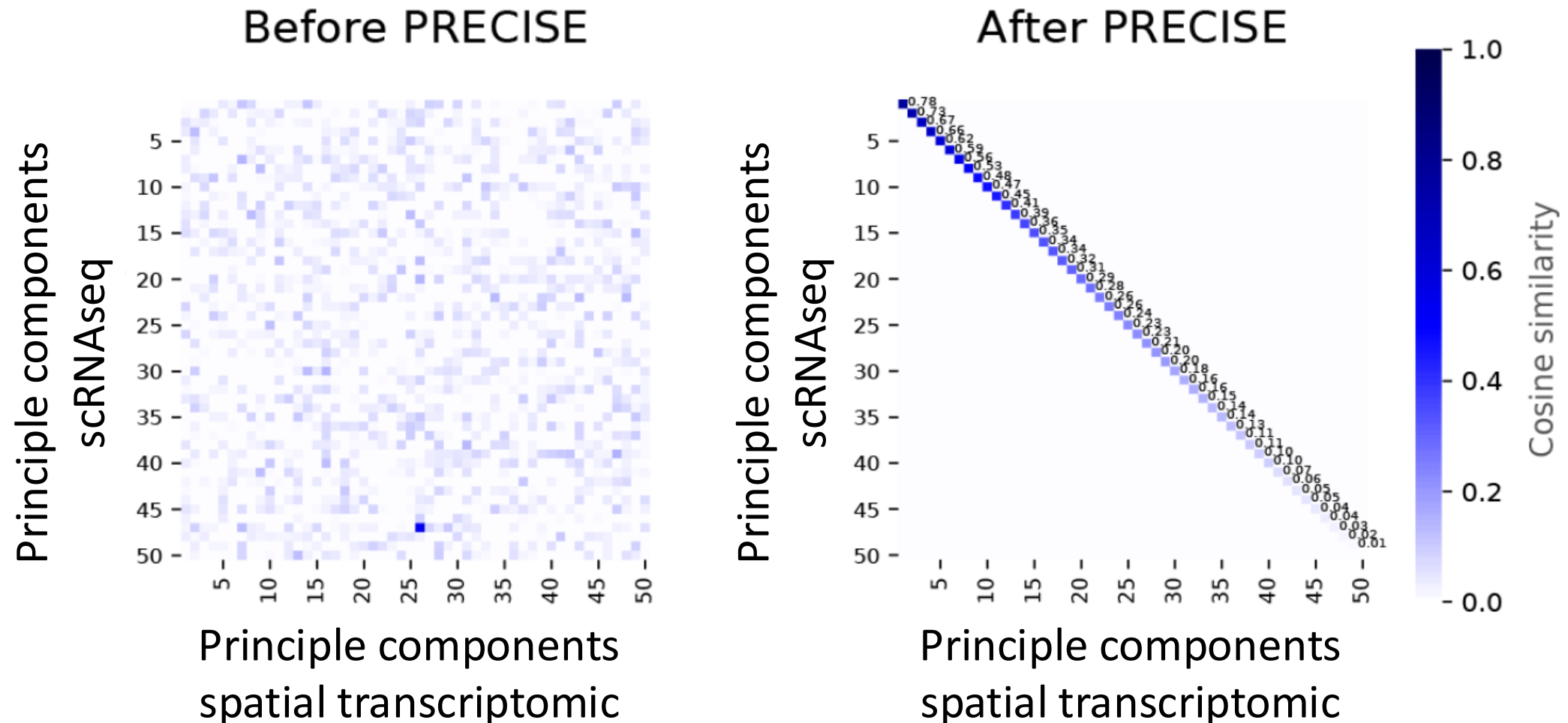


Aligning single-cell and spatial data

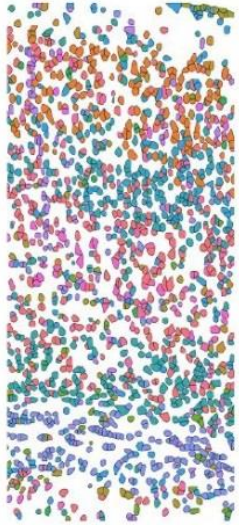
Domain Adaptation using PRECISE



Aligning single-cell and spatial data



SpaGE in primary visual cortex (VISp)

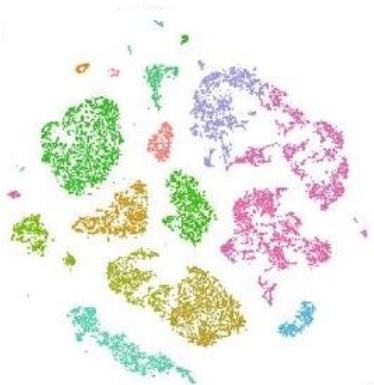


STARmap

1,549 cells

1,020 genes

Wang et al. Science 2018



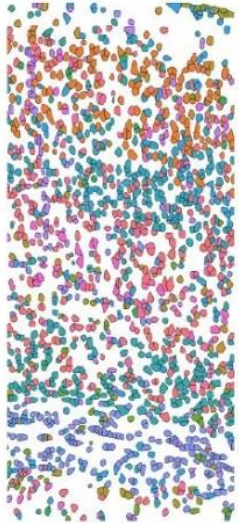
scRNA-seq

14,249 cells

34,617 transcripts

Tasic et al. Nature 2018

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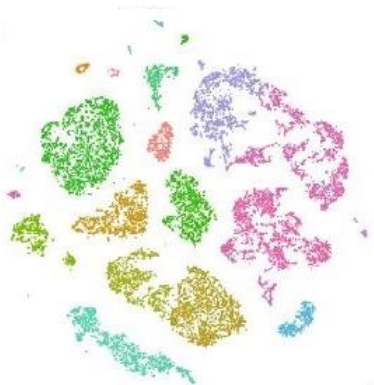
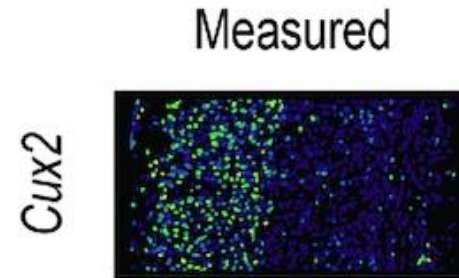


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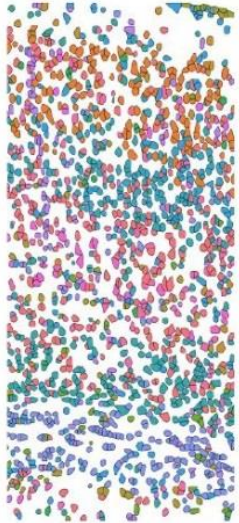
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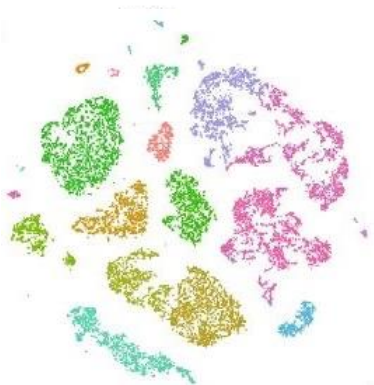
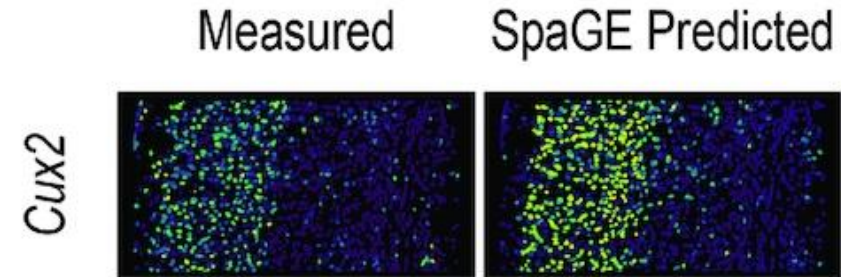


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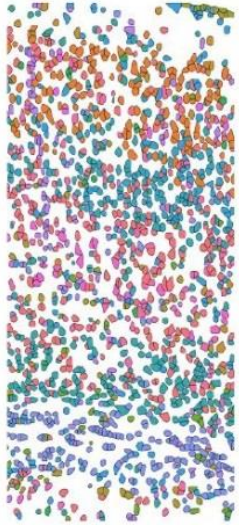
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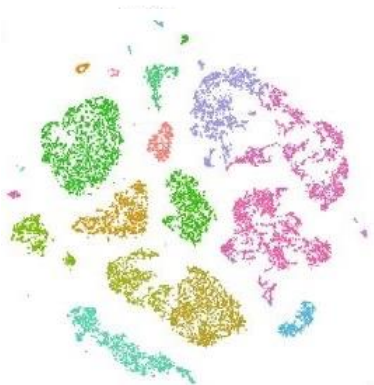


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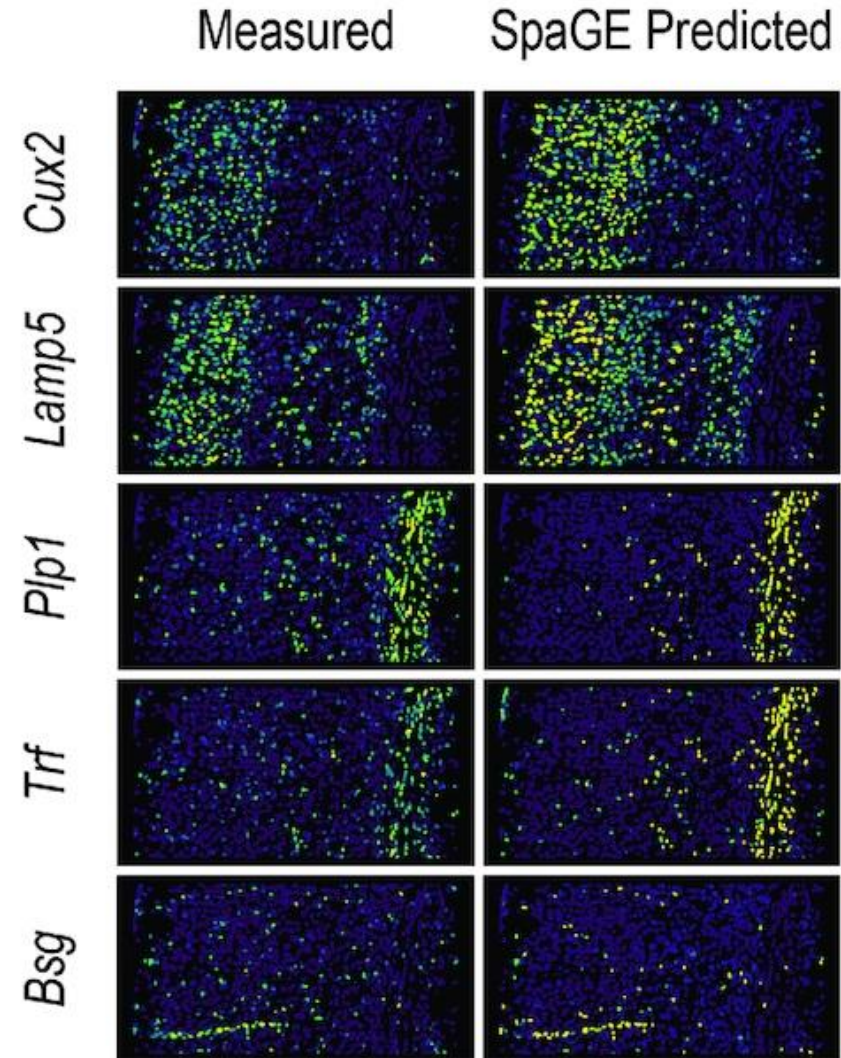


scRNA-seq

14,249 cells

34,617 transcripts

Tasic et al. Nature 2018



How do all these methods compare to each other?

nature communications



Article

<https://doi.org/10.1038/s41467-023-37168-7>

A comprehensive benchmarking with practical guidelines for cellular deconvolution of spatial transcriptomics

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Published online: 21 March 2023

Check for updates

Haoyang Li^{1,2,6}, Juexiao Zhou^{1,2,6}, Zhongxiao Li^{1,2}, Siyuan Chen^{1,2}, Xingyu Liao^{1,2}, Bin Zhang^{1,2}, Ruochi Zhang³, Yu Wang³, Shiwei Sun^{4,5} & Xin Gao^{1,2}✉

Spatial transcriptomics technologies are used to profile transcriptomes while preserving spatial information, which enables high-resolution characterization of transcriptional patterns and reconstruction of tissue architecture. Due to the existence of low-resolution spots in recent spatial transcriptomics technologies, uncovering cellular heterogeneity is crucial for disentangling the spatial patterns of cell types, and many related methods have been proposed. Here, we benchmark 18 existing methods resolving a cellular deconvolution task with 50 real-world and simulated datasets by evaluating the accuracy, robustness, and usability of the methods. We compare these methods comprehensively using different metrics, resolutions, spatial transcriptomics technologies, spot numbers, and gene numbers. **In terms of performance, CARD, Cell2location, and Tangram are the best methods for conducting the cellular deconvolution task.** To refine our comparative results, we provide decision-tree-style guidelines and recommendations for method selection and their additional features, which will help users easily choose the best method for fulfilling their concerns.

Li*, Zhou*..., Gao(2022)

ANALYSIS

<https://doi.org/10.1038/s41592-022-01480-9>

nature | methods

Check for updates

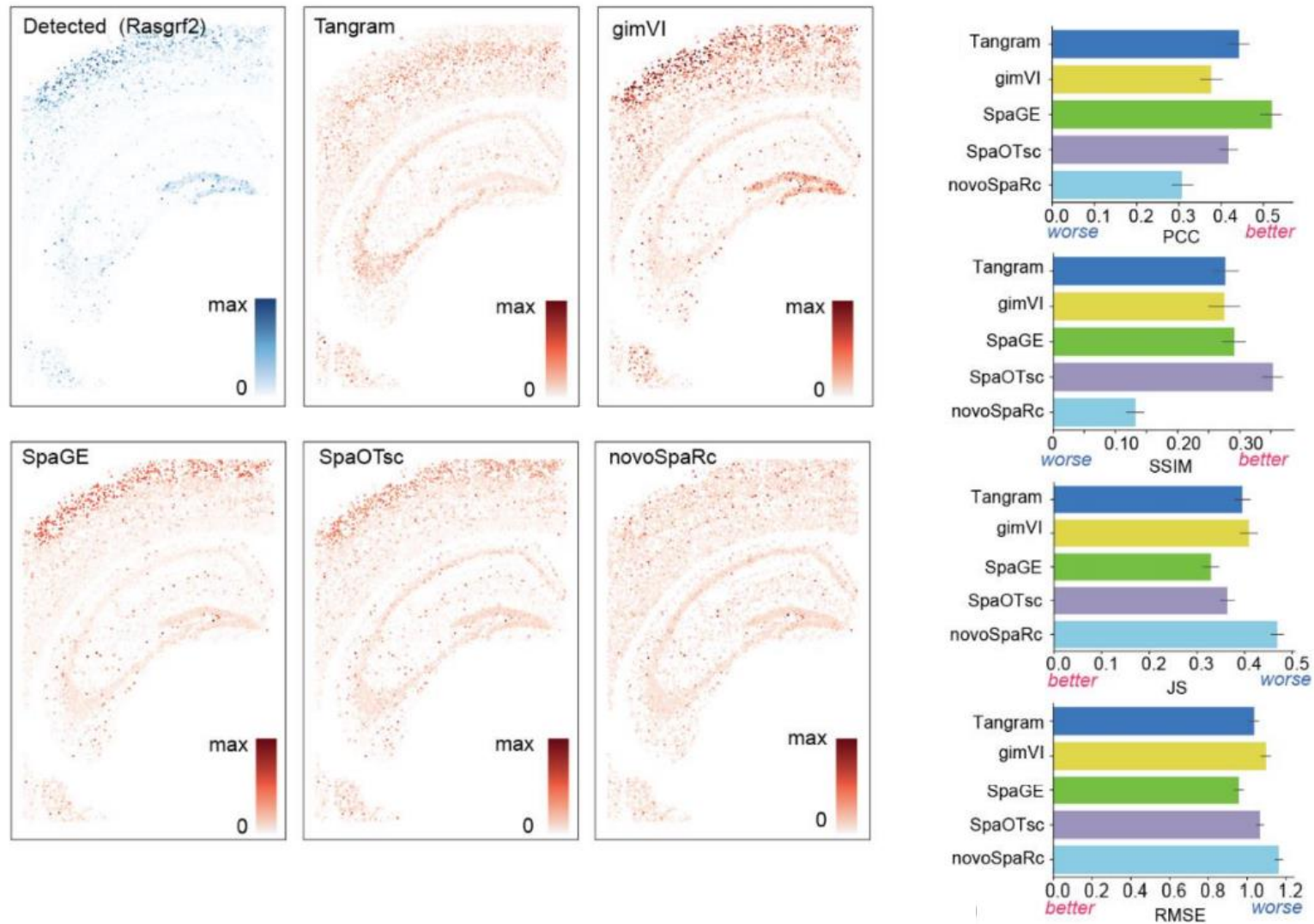
Benchmarking spatial and single-cell transcriptomics integration methods for transcript distribution prediction and cell type deconvolution

Bin Li^{1,7}, Wen Zhang^{1,2,7}, Chuang Guo^{1,7}, Hao Xu^{1,2}, Longfei Li³, Minghao Fang³, Yinlei Hu⁴, Xinye Zhang³, Xinfeng Yao¹, Meifang Tang¹, Ke Liu¹, Xuotong Zhao⁵, Jun Lin^{1,2}, Linzhao Cheng³, Falai Chen⁴, Tian Xue³ and Kun Qu^{1,2,6}✉

Spatial transcriptomics approaches have substantially advanced our capacity to detect the spatial distribution of RNA transcripts in tissues, yet it remains challenging to characterize whole-transcriptome-level data for single cells in space. Addressing this need, researchers have developed integration methods to combine spatial transcriptomic data with single-cell RNA-seq data to predict the spatial distribution of undetected transcripts and/or perform cell type deconvolution of spots in histological sections. However, to date, no independent studies have comparatively analyzed these integration methods to benchmark their performance. Here we present benchmarking of 16 integration methods using 45 paired datasets (comprising both spatial transcriptomics and scRNA-seq data) and 32 simulated datasets. **We found that Tangram, gimVI, and SpaGE outperformed other integration methods for predicting the spatial distribution of RNA transcripts, whereas Cell2location, SpatialDWLS, and RCTD are the top-performing methods for the cell type deconvolution of spots.** We provide a benchmark pipeline to help researchers select optimal integration methods to process their datasets.

Li*, Guo*,..., Qu (2022)

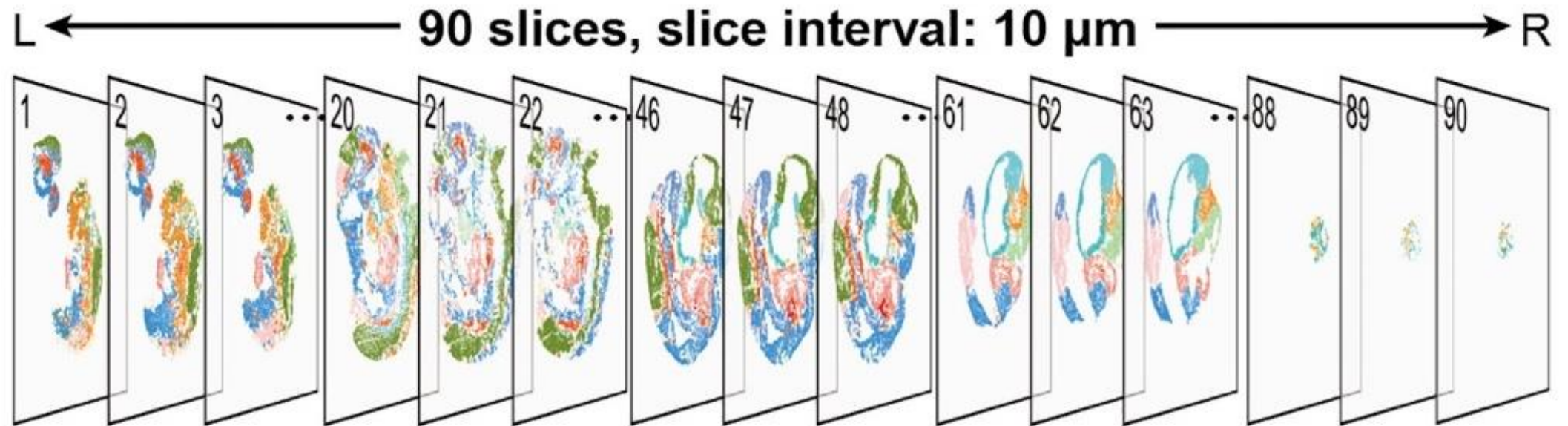
Imputation performance on Xenium data



SRT data alignment

Pseudo 3D / virtual blocks / ...

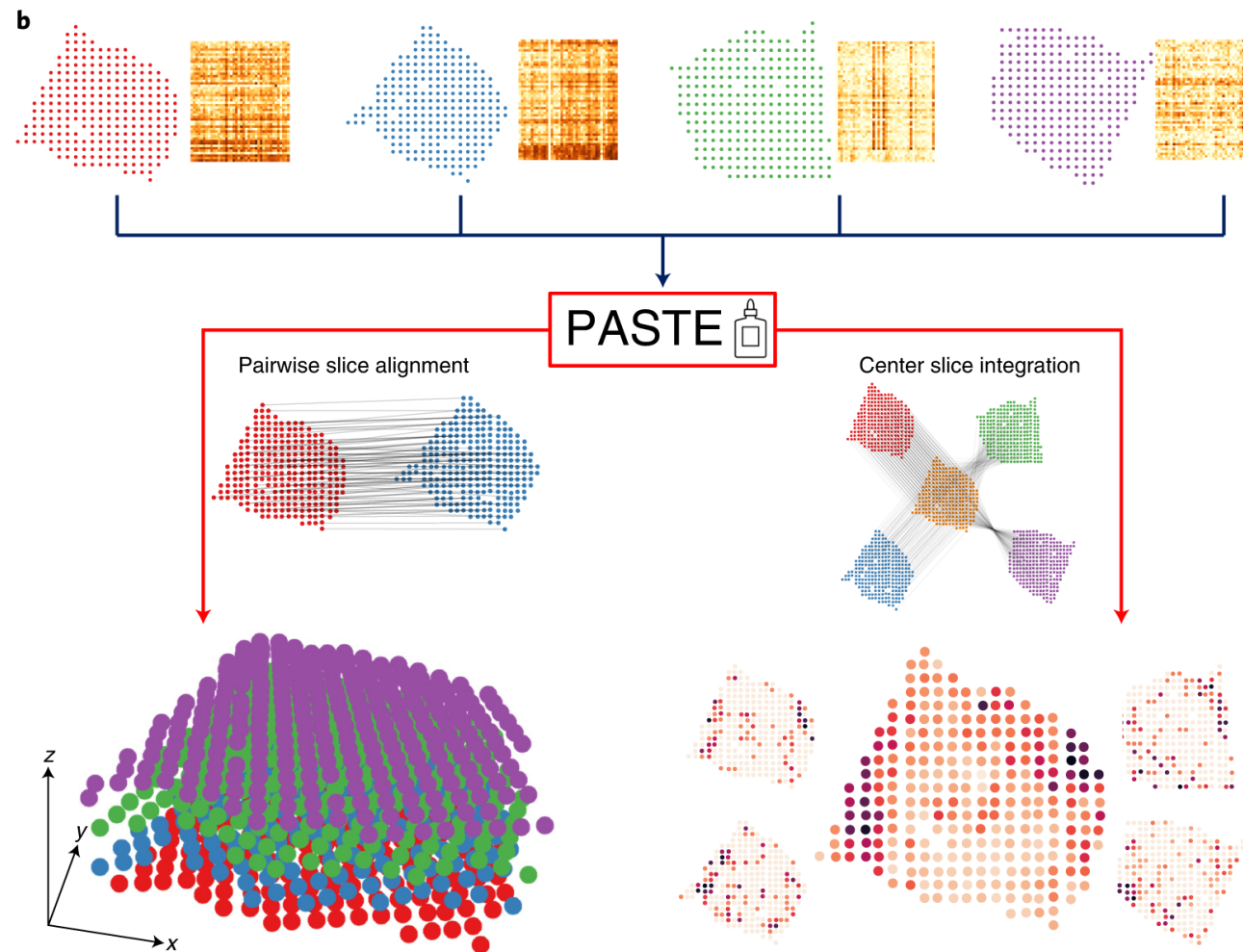
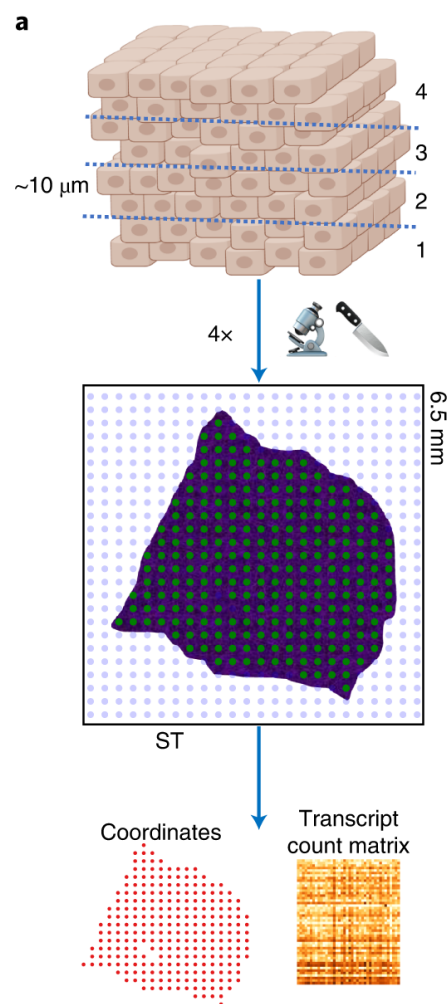
Scalable SRT allows whole tissue mapping using consecutive sections



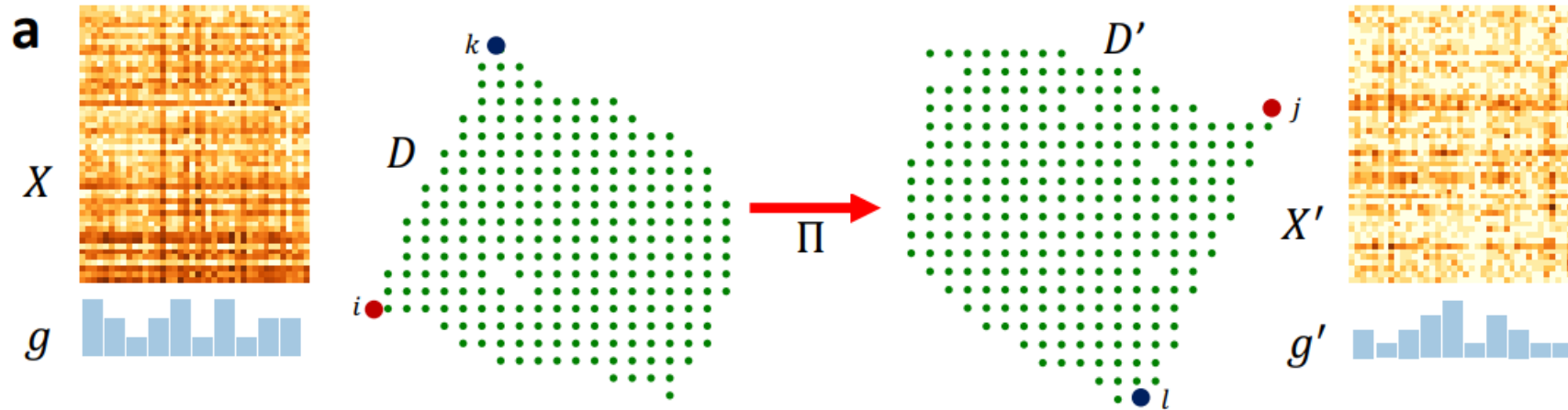
SRT alignment approaches

- Alignment methods: designed to align or match spots or cells from different ST sections or datasets to a common spatial or anatomical reference
 - e.g. PASTE, PASTE2, SPACEL, STalign, GPSA, STIM, CAST
- Integration methods: learn shared latent spot embeddings
 - STAligner, DeepST, PRECAST, SPIRAL

PASTE



PASTE

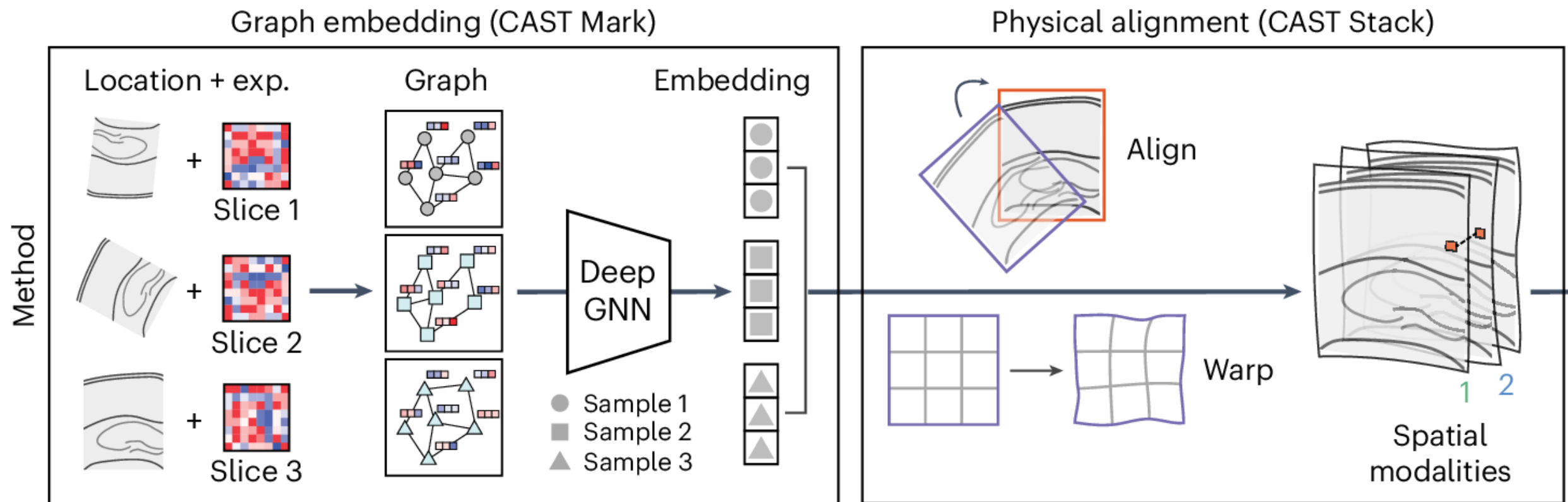


$$F(\Pi; X, D, X', D', c, \alpha) = (1 - \alpha) \underbrace{\sum_{i,j} c(x_{\cdot i}, x'_{\cdot j}) \pi_{ij}}_{\text{Gene expression similarity}} + \alpha \underbrace{\sum_{i,j,k,l} (d_{ik} - d'_{jl})^2 \pi_{ij} \pi_{kl}}_{\text{Spatial distance preservation}}.$$

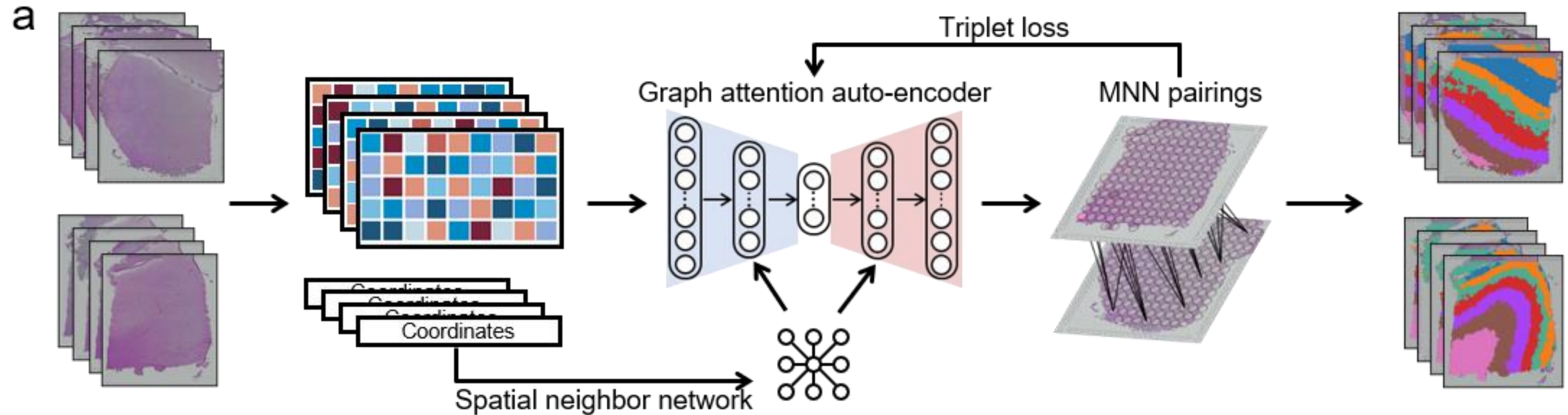
Gene expression
similarity

Spatial distance
preservation

CAST



STAligner

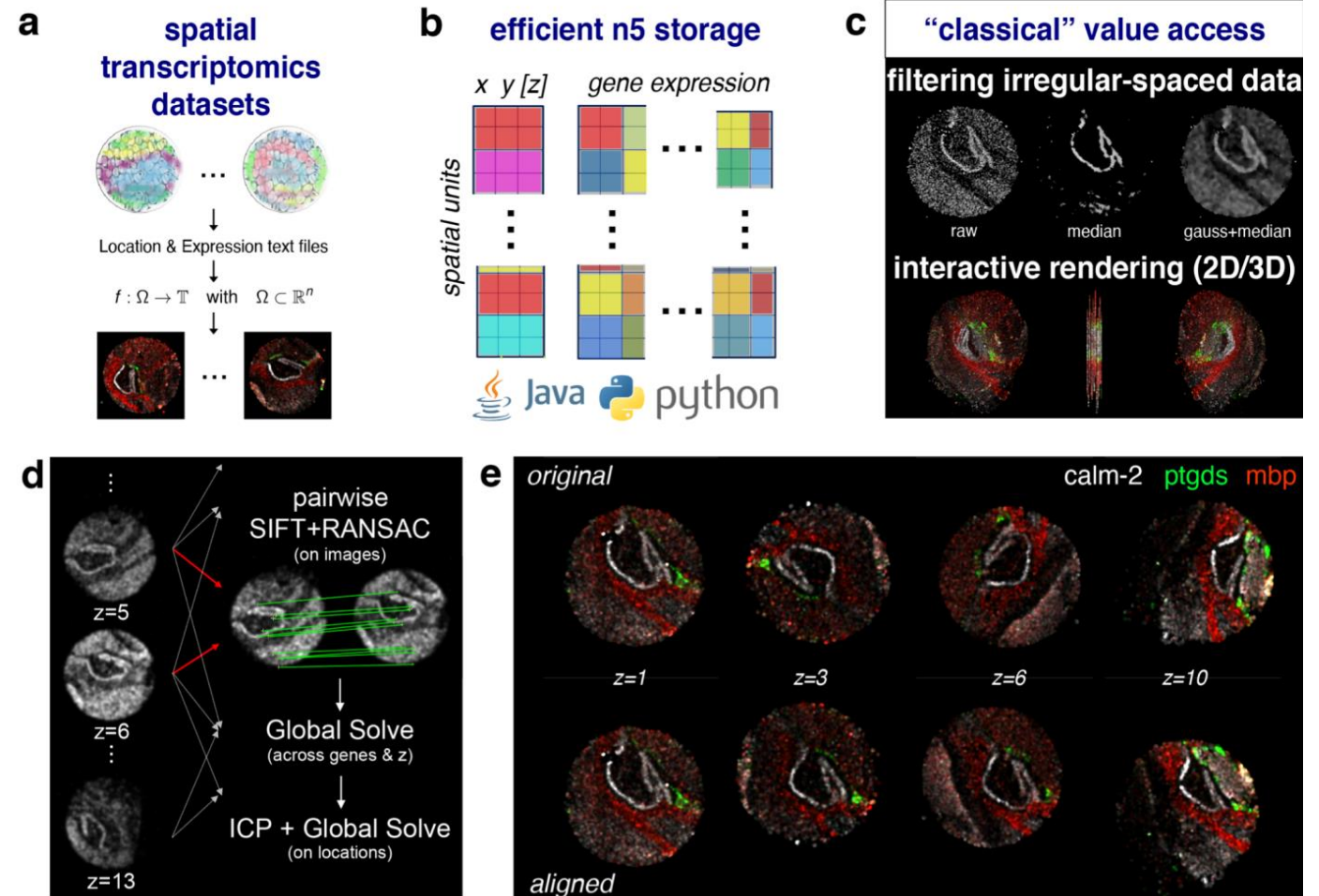
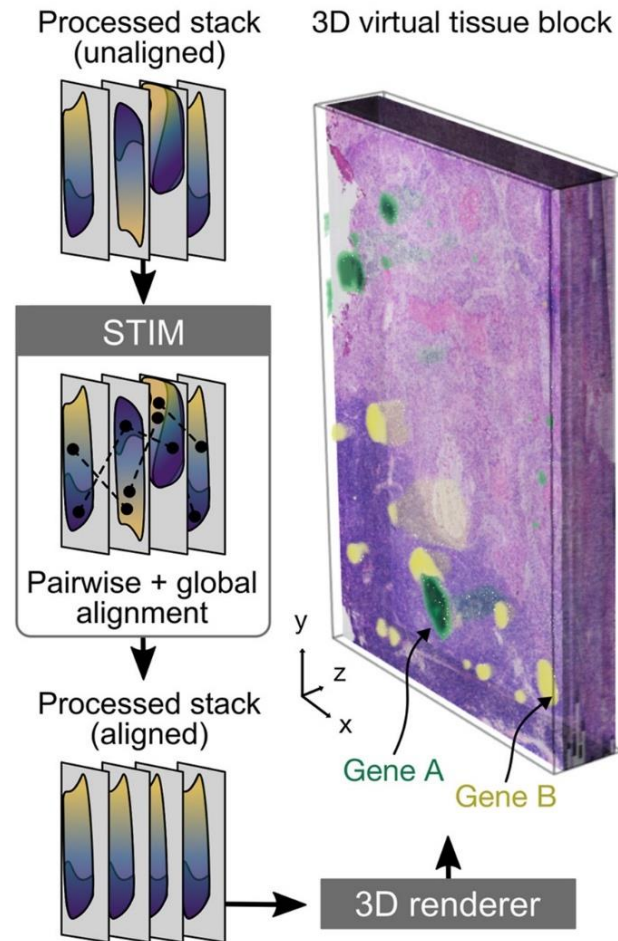


Triplet: anchor-positive and anchor-negative spot pairs

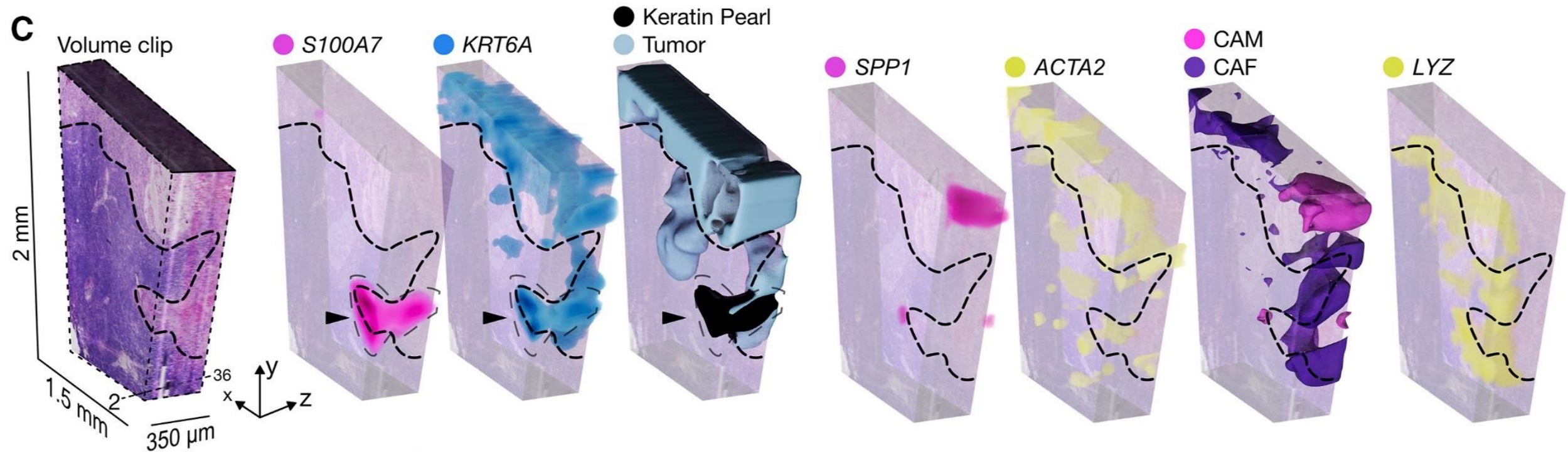
Anchor-positive: mutual nearest neighbors with similar gene expressions but belong to two different slices

Anchor-negative: a pair that belongs to the same slice with different spatial positions and dissimilar expressions

STIM: Spatial Transcriptomics Imaging Framework

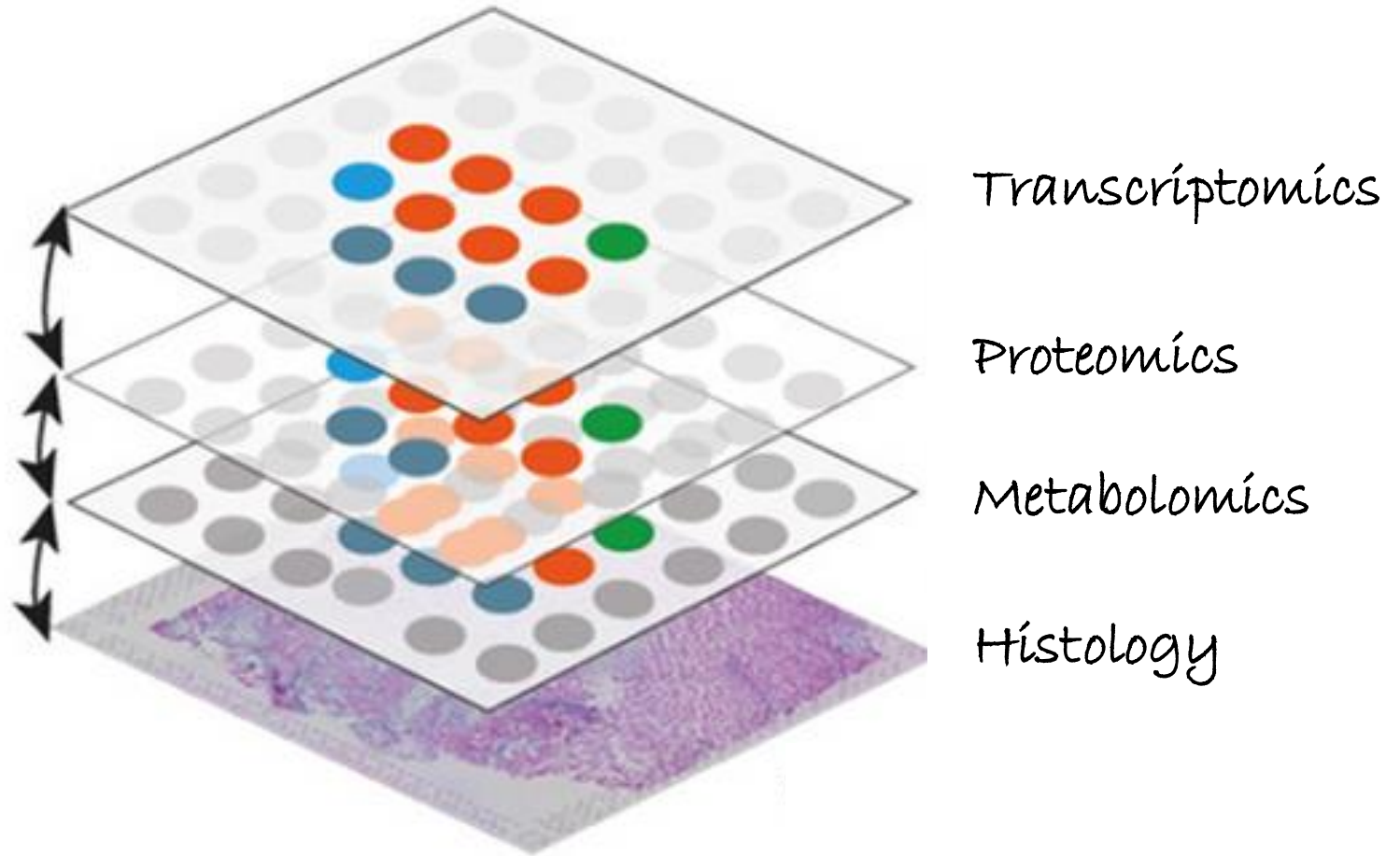


3D information is important

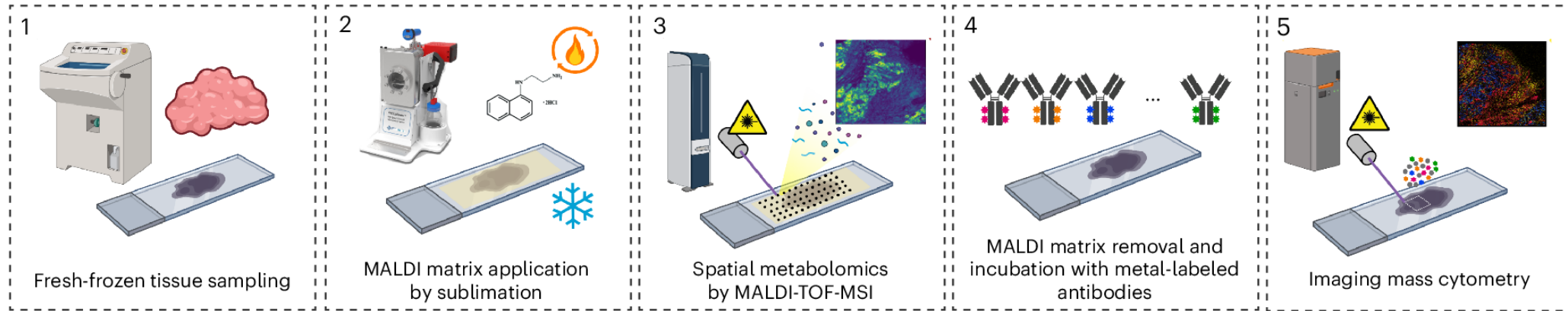


Spatial multiomics

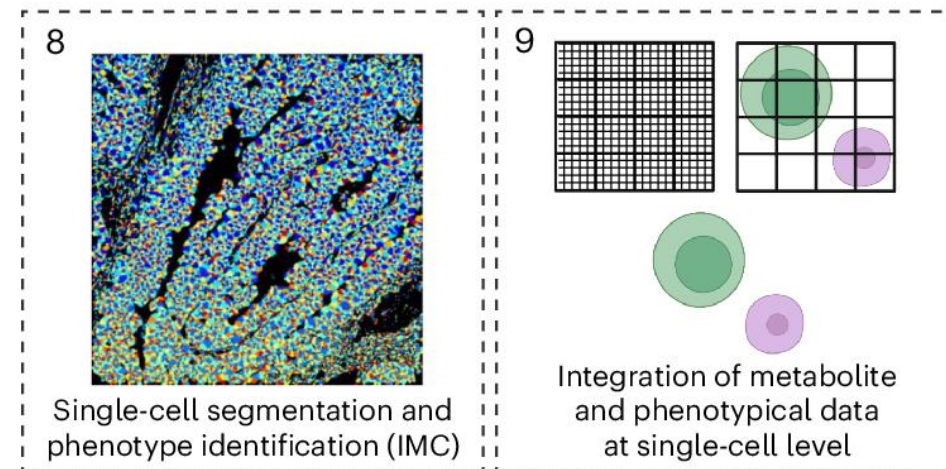
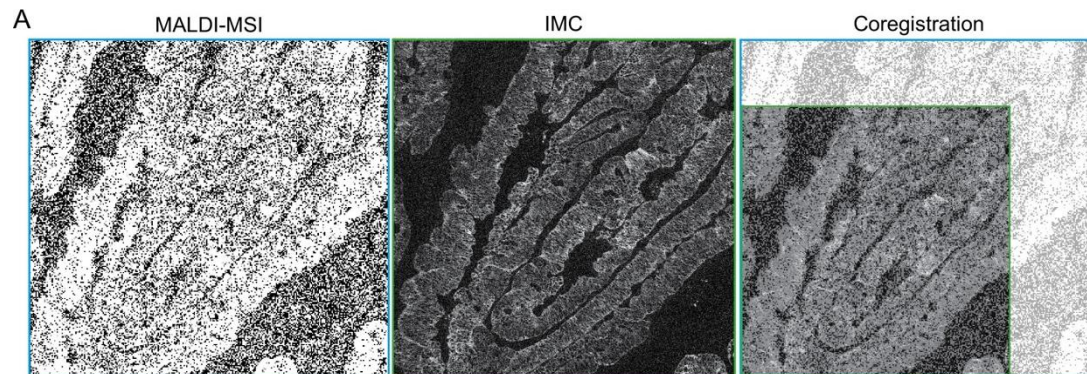
Spatial multimodal



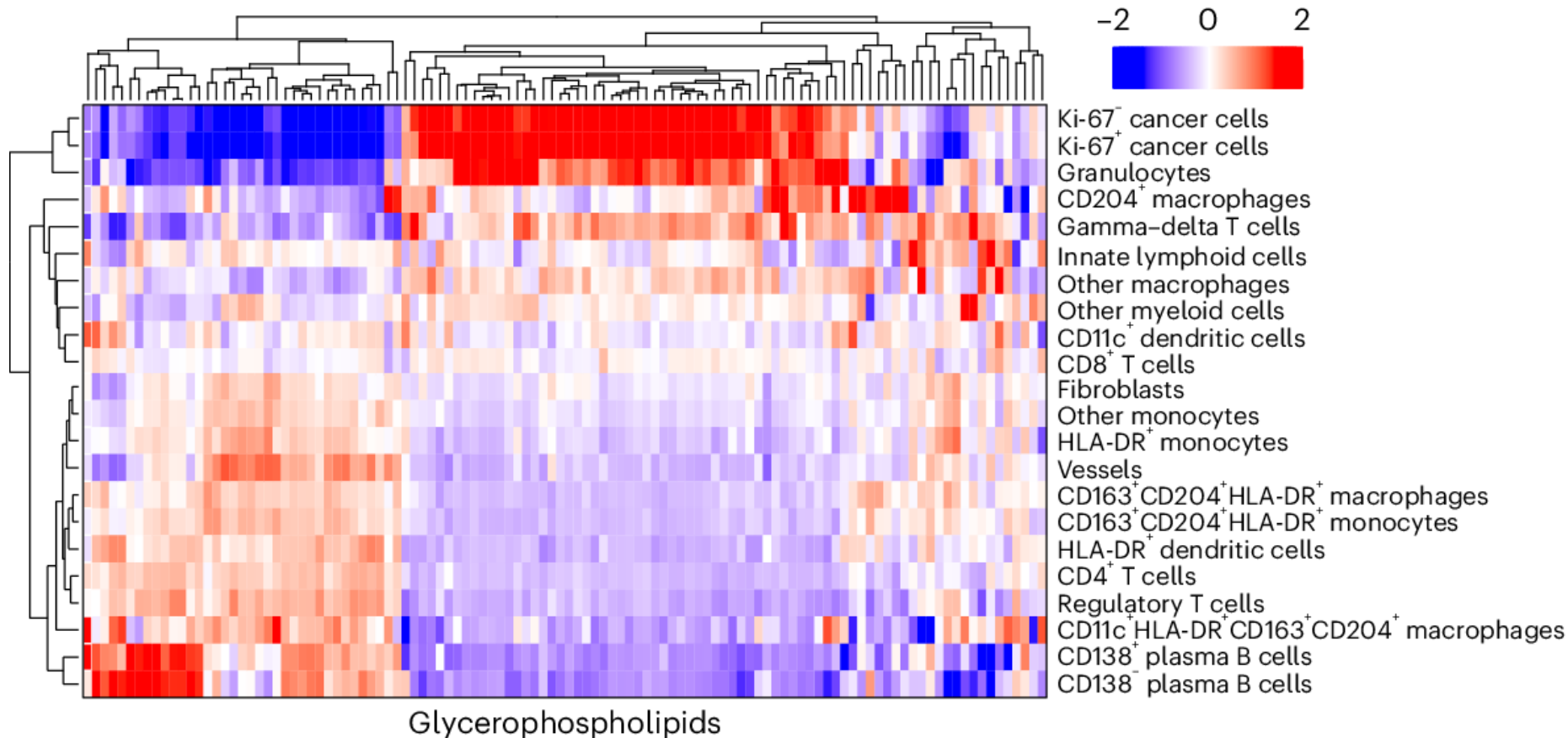
Mass cytometry and mass spectrometry on the same section



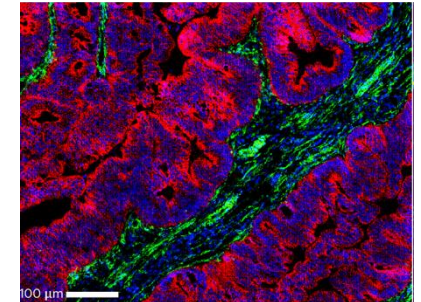
MSI on IMC image alignment



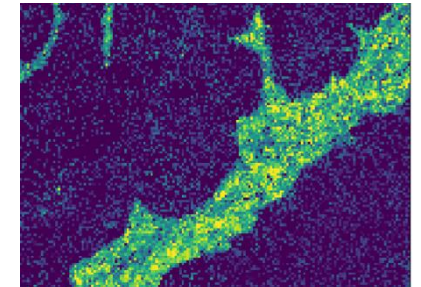
Varying levels of glycerophospholipids across cell types



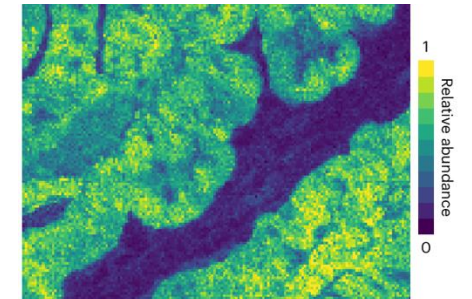
Keratin Vimentin DNA



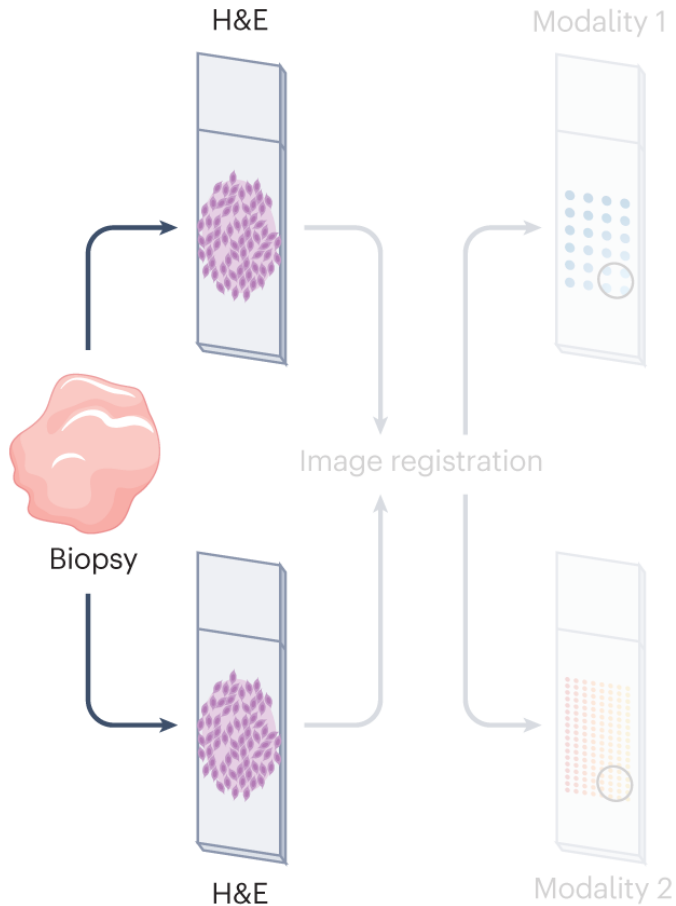
Phosphatidylcholine PC(37:5)



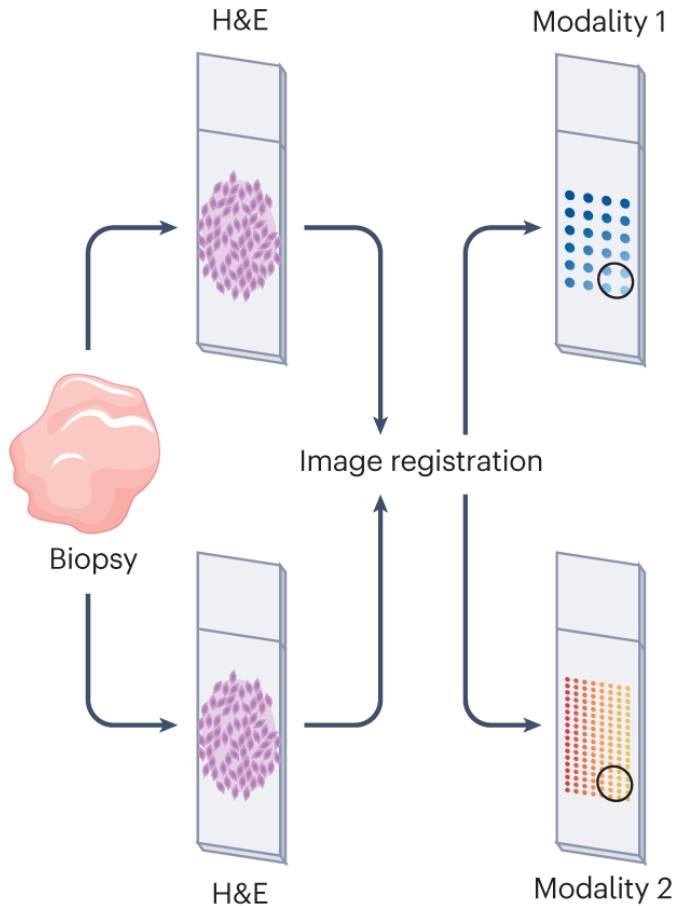
Phosphatidylinositol PI(34:1)



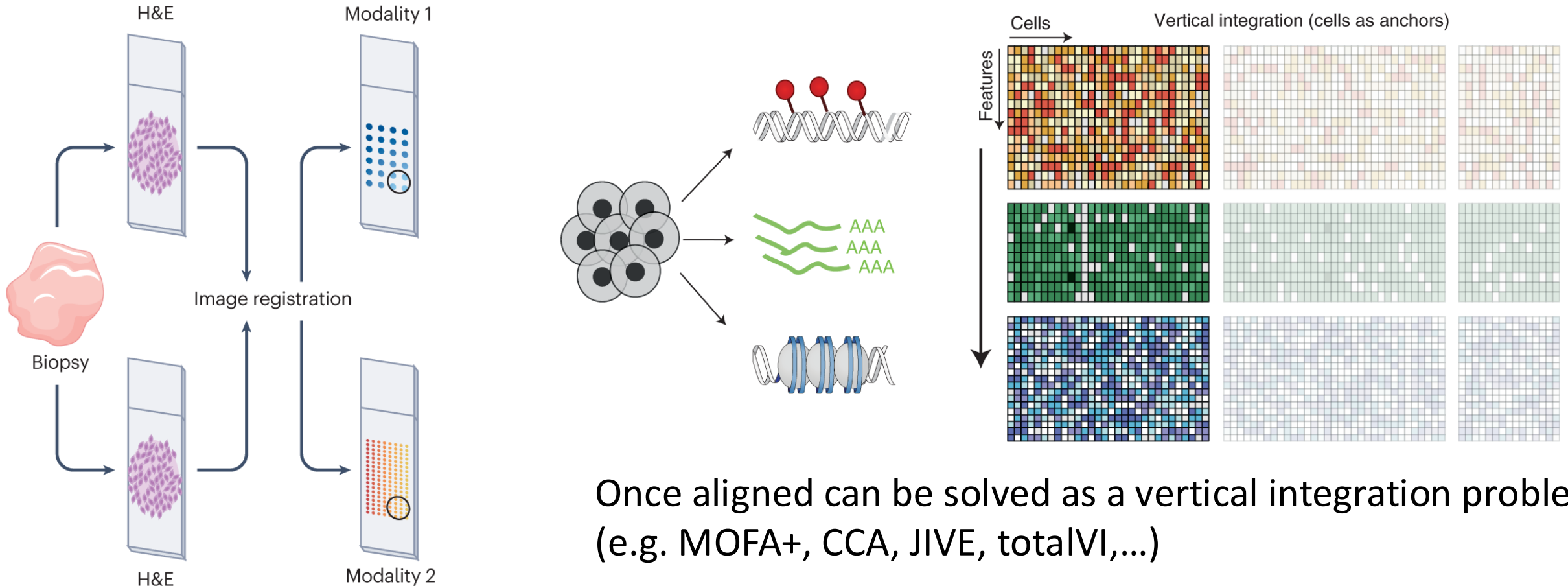
Spatial multiomics on consecutive sections



Spatial multiomics on consecutive sections

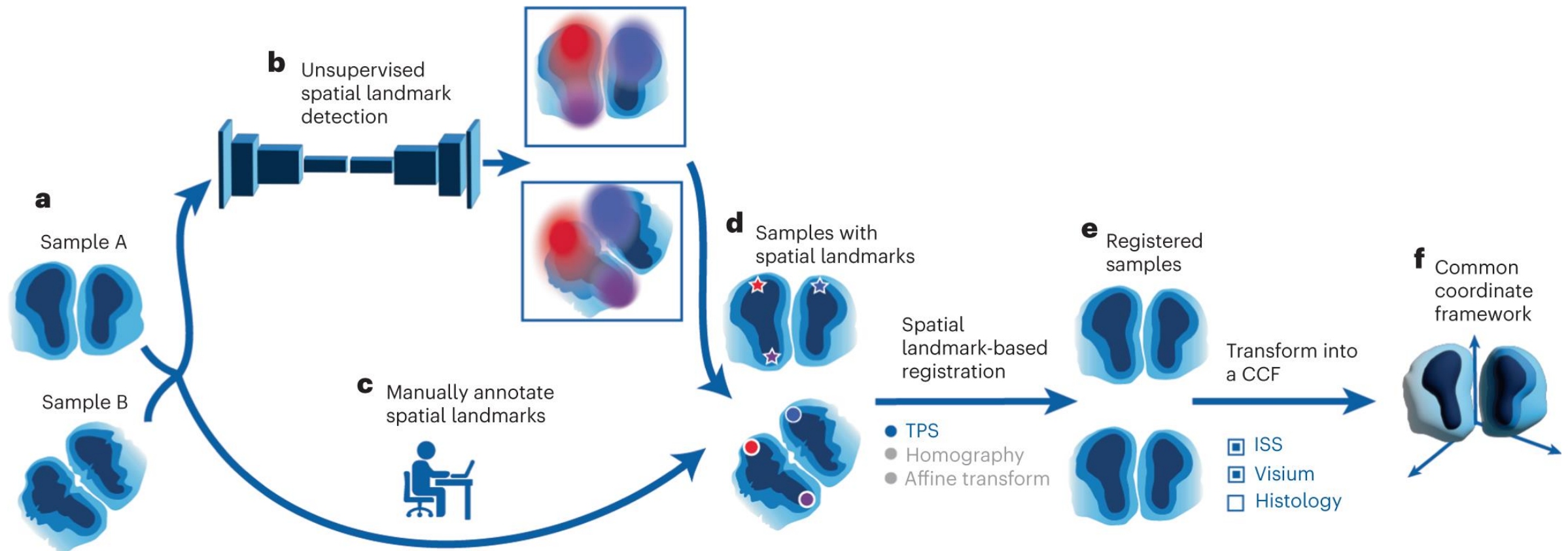


Spatial multiomics on consecutive sections



Once aligned can be solved as a vertical integration problem (e.g. MOFA+, CCA, JIVE, totalVI,...)

Spatial multimodal alignment using Effortless Landmark Detection (ELD)



Summary

- Integrating SRT and sc/snRNA-seq data
- SRT data alignment (pseudo 3D, virtual block,...)
- Spatial multiomics

Thank You!



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